Protein Circuit Topology Plugin

User Manual & Documentation

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Abstract

This is the official documentation file for the PyMOL Protein Circuit Topology plugin. The plugin wraps most of the functionality of the ProteinCT framework [1] and exposes it through a PyMOL GUI. For questions, suggestions, or bug reports, please contact us via email above. The latest version is available at GitHub repository.

Circuit Topology in one paragraph. Circuit Topology (CT) [2] describes how pairs of intramolecular residue–residue contacts are arranged along the sequence. The plugin exposes the four relations also returned by ProteinCT's `local_ct`: **Series (S)** — sequentially arranged, non-overlapping contacts; **Parallel(P)** — one contact nested inside the other; **Inverse Parallel (IP)** — inverse nesting; and **Cross (X)** — interleaved or linked contacts. A *contact map* is the set of residue pairs within a distance cutoff; the plugin renders it as a residue × residue matrix or as an arc ("circuit") diagram above the sequence. In a *relations matrix*, every contact-pair cell is coloured by its CT relation.

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1 Installation Guide

1.1 Downloading and Installing PyMOL

Download the PyMOL 3.1.6.1 installer for your platform (Windows / Linux / macOS) from `https://www.pymol.org/`. Previous versions are available at:

`https://storage.googleapis.com/pymol-storage/installers/index.html`

During setup, install PyMOL for the current user only by selecting **Just Me (recommended)**. This ensures the plugin's automatic dependency installer has the necessary write permissions.

When PyMOL prompts you to register file extensions, select **Register Recommended**. The plugin primarily works with `.pdb`, `.cif`, and `.xtc` files.

1.2 Mandatory Dependencies

The plugin requires the following dependencies to function correctly:

Dependency	Version	Dependency	Version
Python	3.10.18	NumPy	1.26.4
PyQt	5.15.11	Pandas	2.3.3
Qt-Main	5.15.15	Matplotlib	3.9.1
PyQt5-sip	12.17.0	PyMOL	3.1.6.1
Biopython	1.85	PyMOL-Bundle	3.1.6.1

Table 1: Required Python and PyMOL dependencies.

If you follow the installation guide above, only `conda-forge::pandas=2.3.3` and `conda-forge::matplotlib=3.9.1` will require additional installation; everything else ships with the PyMOL bundle.

1.3 Plugin Activation Steps

1. Download the latest plugin release `.zip` directly from GitHub:

   `protein-circuit-topology-pymol-plugin.zip`

   For a specific tagged version, browse the Releases page.

2. Open PyMOL and navigate to **Plugin → Plugin Manager**.

3. Switch to the **Install New Plugin** tab and click **Choose file...**, then select the `.zip` you just downloaded.

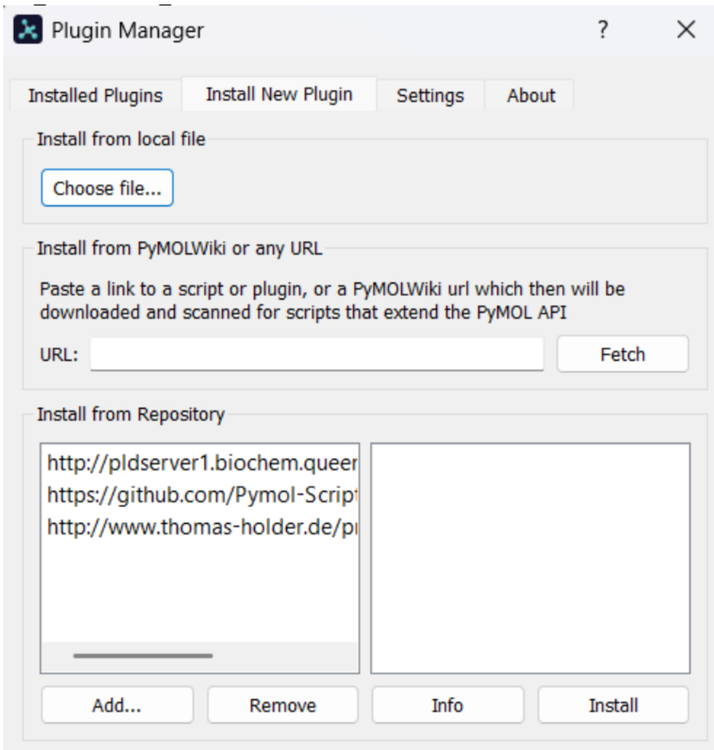

Figure 1: PyMOL's Plugin Manager with the **Install New Plugin** tab open. Use the **Choose file. . .** button under *Install from local file* to select the downloaded `.zip`.

4. Wait for the plugin to install and configure its dependencies. After installation, launch the plugin from **Plugin** → **Protein Circuit Topology Plugin**.

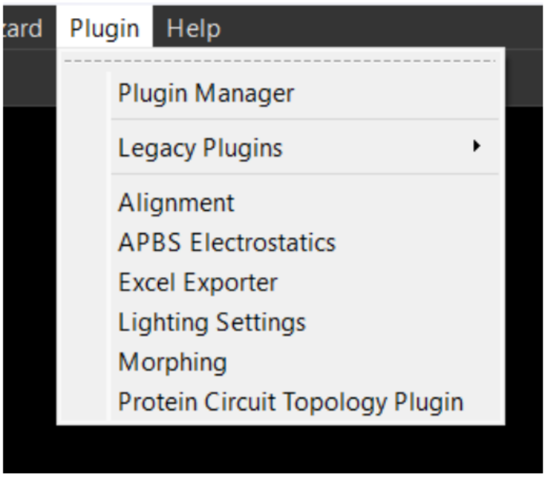

Figure 2: The plugin appears under the PyMOL **Plugin** menu after installation.

Important Installation Notes

- **Initial setup:** When the plugin is imported for the first time, PyMOL may need additional time to download and configure its dependencies. This happens automatically when PyMOL was installed via the official installer.
- **Conda users:** If PyMOL is installed via Conda, the automatic dependency installation through the Plugin Manager may fail. In that case, install the plugin manually and resolve Python dependencies explicitly using the snippet below.

Manual Conda Installation (if applicable)

If you are using a Conda environment and the Plugin Manager fails to install dependencies, run the following command in your terminal before activating the plugin:

```
conda env update --file requirements.yml
```

2 User Guide

2.1 Local Circuit Topology (LCT) Protocol

The Local Circuit Topology (LCT) tab analyses topological relationships within a targeted protein region.

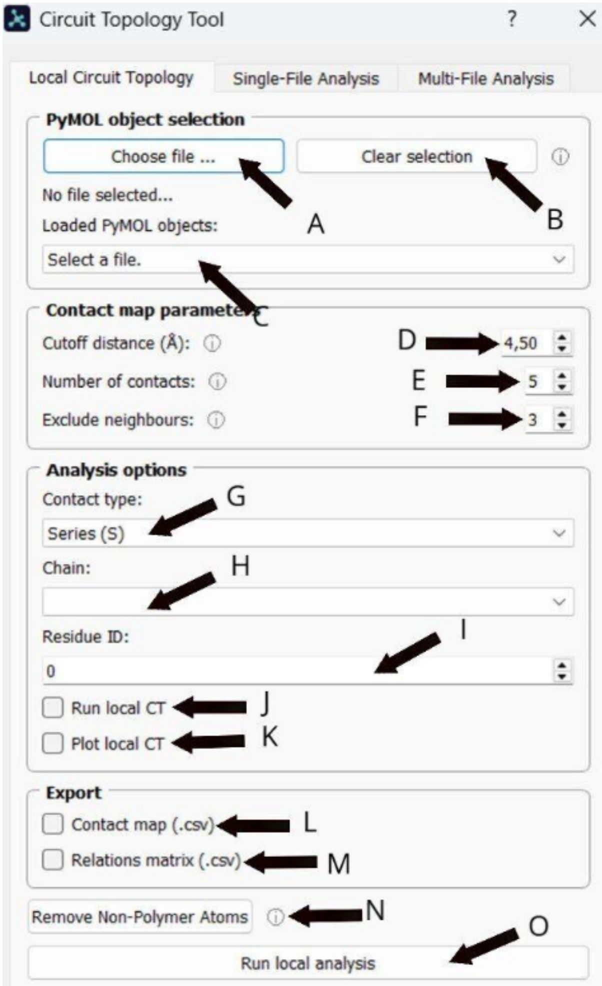

Figure 3: The LCT tab GUI with the controls used in this protocol labelled A–O: file selection (A–B), object selection (C), contact-map parameters (D cutoff distance, E number of contacts, F excluded neighbours), analysis options (G contact type, H chain, I residue ID, J Run local CT, K Plot local CT), exports (L contact map, M relations matrix), structure cleanup (N), and the run button (O).

To narrow the analysis to a specific region, users select a chain (H) and a residue ID (I), then choose the contact type (G) of interest. By checking **Run Local CT** (J), the plugin computes, for every contact that begins or ends at the selected residue, how it relates topologically to every other contact in the structure. Results can be exported as `.csv` files via (L) and (M). For accurate calculations, click **Remove Non-Polymer Atoms** (N) first to exclude waters, ions, and ligands.

LCT Step-by-Step Guide

1. **Load a structure:** Click **Choose file...** (A) to upload a `.pdb` or `.cif` file, or select an already-loaded PyMOL object from the dropdown. Use **Clear selection** (B) to unload it.

2. **Clean the structure:** Click **Remove Non-Polymer Atoms** (N) to drop water molecules, ions, or ligands.

3. **Define contact parameters (D–F):**
   - **Cutoff distance (Å)** — maximum atom–atom distance counted as a contact.
   - **Number of contacts** — minimum atom–atom contacts required to define a residue–residue contact.
   - **Excluded neighbours** — filters out trivial contacts between residues close in sequence.

4. **Target a region:** Pick the **Chain** (H) and **Residue ID** (I).

5. **Choose Contact Type (G):** Series (S), Parallel (P), Inverse Parallel (P -), or Cross (X).

6. **Output options:**
   - Check **Run Local CT** (J) to compute the local topology counts.
   - Check **Plot Local CT** (K) to generate the visualisation shown below.
   - Check **Contact map (.csv)** (L) or **Relations matrix (.csv)** (M) and choose an output directory.

7. **Run:** Click **Run local analysis** (O) to execute.

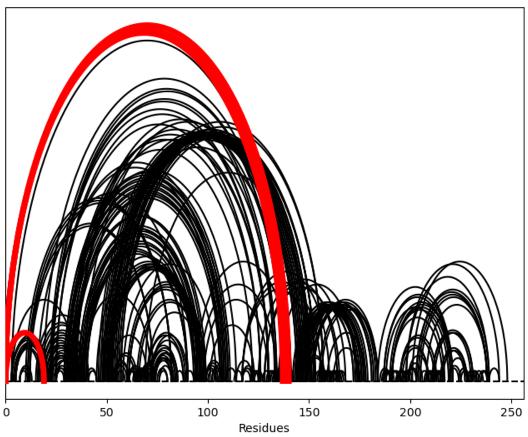

Figure 4: Local CT plot, step 1: contacts that begin or end at the selected residue are highlighted in red over the background contact map.

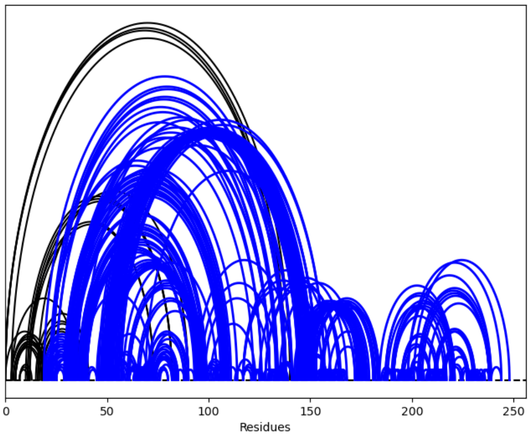

Figure 5: Local CT plot, step 2: contacts whose CT relation to the red ones matches the contact type selected in dropdown (G) are drawn in blue.

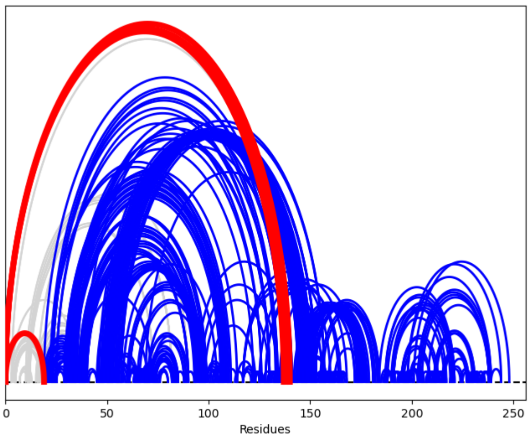

Figure 6: Combined local CT plot showing both the red (residue-anchored) and blue (relation-selected) curves, allowing visual identification of local topological relationships around the selected residue.

2.2 Single-File Analysis Protocol

The Single-File Analysis tab provides a global topological overview of a single protein structure: a contact-map matrix, the corresponding circuit (arc) representation, and an optional scalar **Folding Score** per chain.

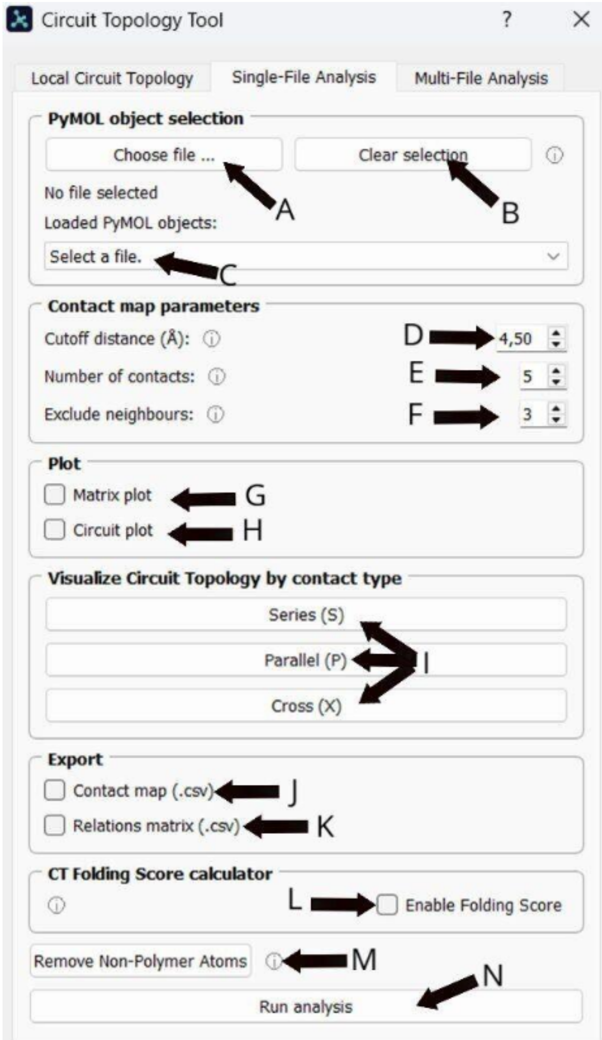

Figure 7: The Single-File Analysis tab. Labels: file/object selection (A–C), contact-map parameters (D–F), plot toggles (G matrix, H circuit), the *Visualize Circuit Topology by contact type* buttons (I row: S/P/X), export checkboxes (J–K), Folding Score (L), structure cleanup (M), and the run button (N).

The **Matrix Plot** renders the relations matrix: each cell colours the CT relation between two contacts. For *single-chain* structures the matrix uses the standard P / S / X categories (with inverse variants). For *multi-chain* structures the plugin adds the L / T / I categories visible in the legend of Figure 9. The **Circuit Plot** draws every contact as an arc above the residue index; for multi-chain inputs each chain is rendered in a different colour (see Figure 8).

Note: When both ***Matrix plot*** *(G) and* ***Circuit plot*** *(H) are selected, the plugin opens two separate windows — one for each plot.*

Single-File Analysis Guide

1. **Load the protein (A–C):** Upload a `.pdb` or `.cif` file or pick an existing PyMOL object from the dropdown.

2. **Clean the structure (M):** Click **Remove Non-Polymer Atoms** if heteroatoms are present.

3. **Define contact parameters (D–F):** Cutoff distance, number of contacts, excluded neighbours — the same parameters as in the LCT tab.

4. **Select plots:**

   - Check **Matrix plot** (G) for the relations matrix.
   - Check **Circuit plot** (H) for the arc diagram.

5. **(Optional) Folding Score:** Check **Enable Folding Score** (L) to compute a per-chain CT FoldScore (see Section 2.2.2).

6. **Export data:** Tick **Contact map (.csv)** (J) and/or **Relations matrix (.csv)** (K) and pick an output directory.

7. **Run:** Click **Run analysis** (N).

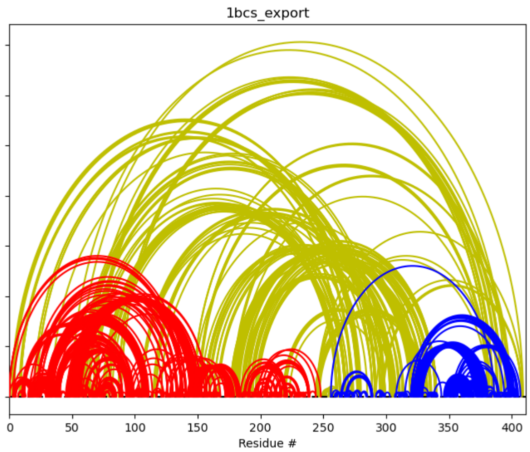

Figure 8: Circuit (arc) plot produced by the Single-File Analysis tab for the multi-chain object `1bcs_export`. Each chain is drawn in a distinct colour; arcs spanning chains highlight inter-chain contacts.

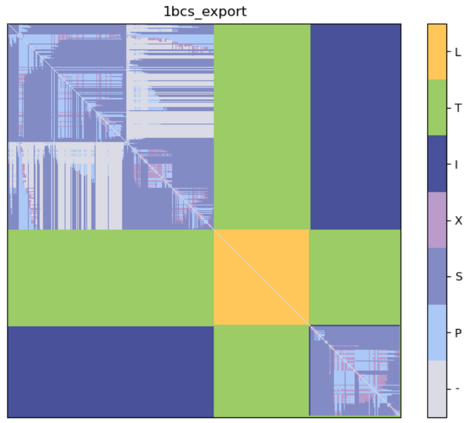

Figure 9: Relations-matrix plot for the same multi-chain object. The legend uses the multi-chain contact categories; the extra L, T, I bins appear only for multi-chain proteins. For single-chain inputs the legend reduces to the standard CT classes (P, S, X).

2.2.1 Visualizing Topology on the 3D Protein Structure in PyMOL

The *Visualize Circuit Topology by contact type* panel (Figure 7, button row I) recolours the loaded molecule directly in PyMOL according to how densely each residue participates in a given CT relation. After running the analysis (so the relations matrix exists), pressing **Series (S)**, **Parallel (P)**, or **Cross (X)** performs two steps internally:

1. `get_topology_vector(mat, index, topology_type, numbering)` — builds a per-residue density vector counting, for that residue, the number of contacts of the selected CT type.

2. `color_by_topology(molecule_name, vector, numbering, topology_type)` — maps the density onto a PyMOL colour gradient and applies it to the selected object.

This makes it possible to map topology descriptors back onto the 3D fold and identify, for example, regions densely involved in cross contacts. See the figures below.

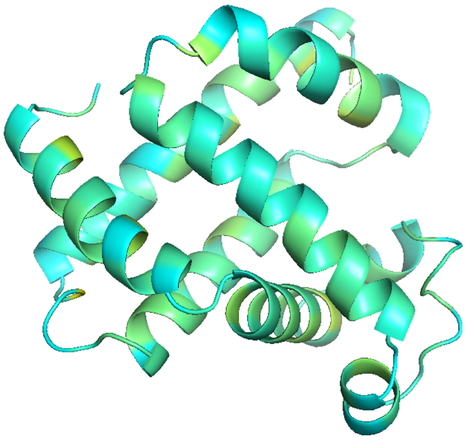

Figure 10: Contact type **Series (S)** visualized for protein `1a6n`. From the command line: '*The minimum number of S contacts in 1a6n for a single residue is 0.000000. It is colored with cyan. The maximum number of S contacts in 1a6n for a single residue is 757.000000. It is colored with yellow.*'

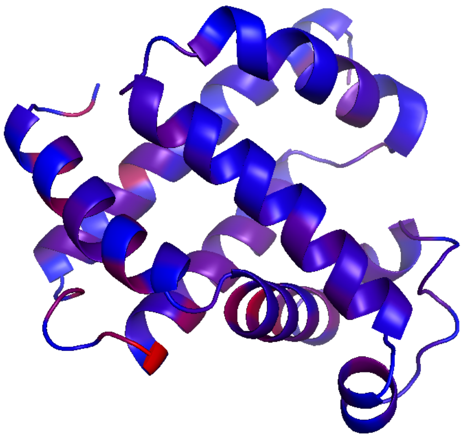

Figure 11: Contact type **Parallel (P)** visualized for protein `1a6n`. From the command line: '*The minimum number of P contacts in 1a6n for a single residue is 0.000000. It is colored with blue.The maximum number of P contacts in 1a6n for a single residue is 396.000000. It is colored with red.*'

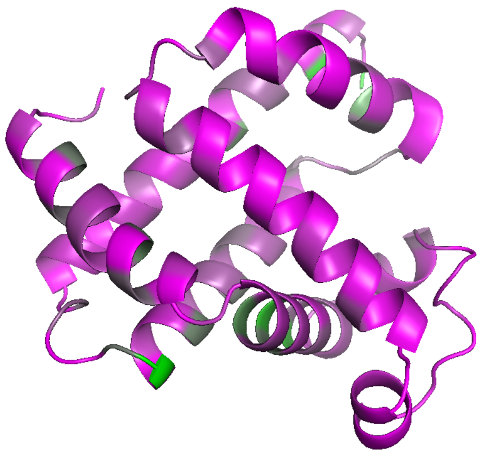

Figure 12: Contact type **Cross (X)** visualized for protein `1a6n`. From the command line: '*The minimum number of X contacts in 1a6n for a single residue is 0.000000. It is colored with magenta. The maximum number of X contacts in 1a6n for a single residue is 151.000000. It is colored with green.*'

2.2.2 CT Folding Score

Ticking **Enable Folding Score** (L) calls `utils/folding_score.py::get_folding_score`, which combines the contact map and the relations matrix into a single scalar per chain, summarising how entangled the contact network is. The score builds on the circuit-topology-based fold analysis framework of Akulov *et al.* [3] and provides a quick way to compare conformations without inspecting full plots. See example output in Figure 13 below.

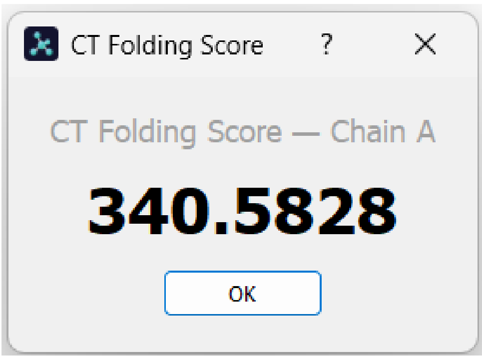

Figure 13: CT Folding Score for protein `1a6n`.

2.3 Multiple-File Analysis Protocol

The Multiple-File Analysis tab batch-processes a set of structures: either a directory of `.pdb`/`.cif` files, or a trajectory loaded as a topology `.pdb` plus an `.xtc` trajectory file.

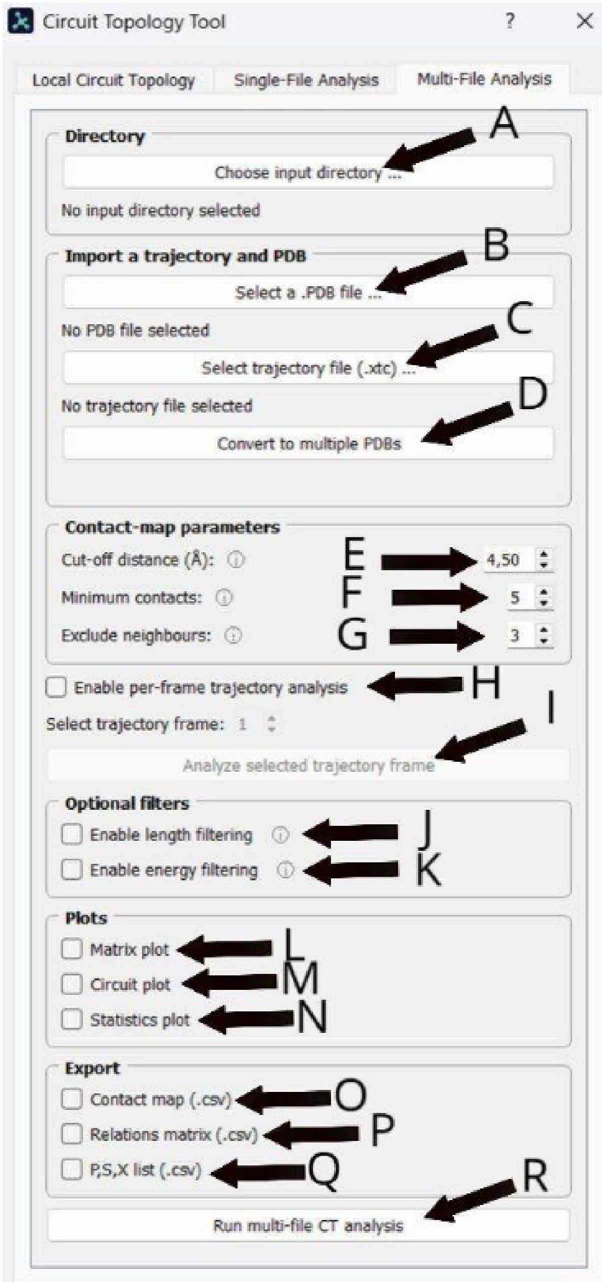

Figure 14: The Multi-File Analysis tab. Labels: directory input (A), trajectory import (B–C), trajectory–to–PDB conversion (D), contact-map parameters (E–G), per-frame analysis (H–I), optional filters (J length, K energy), plot toggles (L matrix, M circuit, N statistics), export checkboxes (O–Q), and the run button (R).

Trajectory Workflow

Two input modes are supported. Either:

- point at a **directory** of structures via **Choose input directory . . .** (A), or

- import a topology+trajectory pair: **Select a .PDB file ...** (B) and **Select trajectory file (.xtc) ...** (C).

With a topology + trajectory loaded you have the additional option to **Convert to multiple PDBs** (D) to export each frame to its own `.pdb` file, which turns the trajectory into a directory you can then point at for a regular multi-file run.

Optional Filters

Two new filter groups are available on this tab:

- **Length filter** (J): keeps only contacts whose sequence separation matches the chosen comparator (`<`, `=`, `>`) against a distance threshold.
- **Energy filter** (K): retains only attractive (stabilising) or only repulsive (destabilising) contacts.

Filter caveats

The length filter and energy filter were designed for single-chain proteins and are skipped automatically when the input contains multiple chains. They also have no effect during *per-frame trajectory analysis*, which runs the unfiltered pipeline on the chosen frame. Aggressive length-filter thresholds can produce empty contact maps.

Plots and Exports

Three plot checkboxes are available:

- **Matrix plot** (L) and **Circuit plot** (M) — the same outputs as the Single-File tab, generated for every input file.
- **Statistics plot** (N) — the fraction of entangled (Parallel + Cross) contacts plotted against the diagonal offset, averaged across the input set.

Four export checkboxes (O–Q) write CSV outputs to the chosen directory: the residue contact map (O), the relations matrix (P), and the per-residue P, S, X counts (Q). Ticking the P,S,X list export enables an extra **P,S,X contacts plotted over time** checkbox, which produces a time-series plot of the three counts across input files — only informative for genuine trajectories (i.e. when each input file is a consecutive frame of the same structure).

Multiple-File Analysis Guide

1. **Load the input:** either pick an input directory (A) of `.pdb`/`.cif` files, or load a topology+trajectory pair (B–C).
2. **(Optional) Convert trajectory to PDBs (D)** if you want to inspect or process individual frames as standalone files.
3. **Define contact parameters (E–G):** Cutoff distance, minimum contacts, excluded neighbours.

4. **(Optional) Per-frame analysis (H–I):** tick **Enable per-frame trajectory analysis** and choose the frame index to inspect.
5. **(Optional) Apply filters (J–K):** enable **Length filter** and/or **Energy filter** subject to the caveats above.
6. **Pick plots (L–N):** Matrix, Circuit, Statistics.
7. **Pick exports (O–Q):** Contact map, relations matrix, P,S,X list. If P,S,X list is selected you can additionally tick **P,S,X contacts plotted over time**.
8. **Run:** Click **Run multi-file CT analysis** (R) to process the whole input set, or **Analyze selected trajectory frame** to process just the chosen frame.

References

[1] Moes, D., Banijamali, E., Sheikhhassani, V., Scalvini, B., Woodard, J., Mashaghi, A. (2022). *ProteinCT: An implementation of the protein circuit topology framework.* MethodsX, open access (PII S2215016122002400). Source code: https://github.com/circuittopology/circuit_topology.

[2] Mashaghi, A., Van Wijk, R.J., Tans, S.J. (2014). *Circuit topology of proteins and nucleic acids.* Structure 22(9), 1227–1237. doi:10.1016/j.str.2014.06.015.

[3] Akulov, V., Panizo, A.J., Estébanez-Perpiñá, E., van Noort, J., Mashaghi, A. (2024). *Phosphorylation regulated conformational diversity and topological dynamics of an intrinsically disordered nuclear receptor.* bioRxiv 2024.09.21.614239, Cold Spring Harbor Laboratory. doi:10.1101/2024.09.21.614239.